

İTÜ

Security in Financial Information Systems

Applied Cryptography

Dr. Şerif Bahtiyar
bahtiyars@itu.edu.tr

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Motivation

The Microsoft Exchange hack shows attackers are working 'smarter, not harder,' experts say



The recent **hack** has received less attention since victims were **small- to mid-size** organizations. But the hack could be a "**test run**" for a **larger attack**, meaning Americans must pay attention.

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Motivation

Elon Musk denies Tesla cars are used for spying in China



Elon Musk, the chief executive of the US electric carmaker **Tesla**, says his firm would be "shut down" if its **vehicles** were used for **spying on China**.

<https://www.bbc.com/news/world-us-canada-56471681>

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Outline

- Basic concepts of cryptography
- Symmetric Cryptosystem
- Substitution Techniques
- Transposition Techniques
- Block cipher
- Random Numbers
- Stream Ciphers
- Principles of Public-key Cryptography
- RSA

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Basic concepts

Cryptography

- **Crypto**: secret, hidden
- **Graph**: writing or study
- **Cryptography** is a study of **secret writing** to ensure **secure systems** in the **presence of adversaries**.



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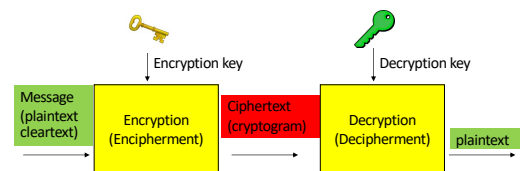
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Basic concepts

Basic scenario



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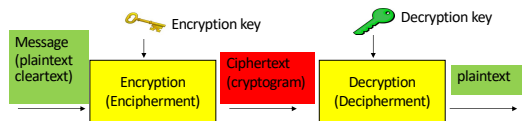
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Basic concepts

- **Plaintext**: the **original** message
- **Ciphertext**: the **scrambled** message
- **Encrypt**: converting **plaintext** to **ciphertext**
- **Cipher**: **algorithm** for transforming plaintext to ciphertext



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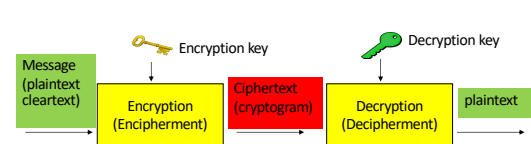
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Basic concepts

- **Decrypt**: recovering **plaintext** from **ciphertext**
- **Key**: information used in cipher **known only to sender/receiver**
- **Secret Key**: a value **independent of the plaintext** and of the **algorithm**.
- **Cryptanalysis**: The **process** of attempting to **discover the plaintext** or **key**
- **Cryptography**: The areas of **cryptography** + **cryptanalysis**



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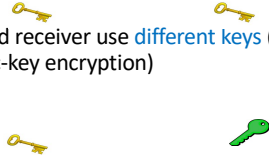
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Classification of Cryptographic Systems

The Number of Keys Used

- Sender and receiver use the **same key** (**symmetric**, single-key, conventional encryption)
- Sender and receiver use **different keys** (asymmetric, two-key, public-key encryption)



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Classification of Cryptographic Systems

Type of Operations (Symmetric-key cryptography)

- It is used for **transforming** plaintext to ciphertext.
- **Substitution (S)** (bit, letter, or group of bits or letters)
İTÜ -> 478 (İ->4, T->7, Ü->8)
- **Transpositions (T)**
İTÜ -> TÜİ (123 -> 231)
- **Product**: **multiple stages** of substitutions and transpositions
STSSST
- **Requirement**: no information be lost!

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Classification of Cryptographic Systems

The way in which the plaintext is processed.

- **Block cipher**

Istanbul -> qwertyuo

- **Stream cipher**

Istanbul -> qstanbul

Istanbul -> qwanbul

.

Istanbul -> qwertyuo

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Symmetric Cryptosystem

There are **two requirements** for secure use of symmetric encryption:

- **Strong encryption** algorithm

*The **opponent** should be **unable** to **decrypt ciphertext** or discover the **key** even with **ciphertexts together** with the **plaintext**.*

- **Secure key Distribution**: Sender and receiver must have **obtained** copies of **secret key** in a **secure** fashion and must **keep** the key **secure**



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Symmetric Cryptosystem

- Generally, it is **assumed** that **opponent**
 - Knows encryption **algorithm**
 - Does **not** know **keys**
- This implies that a **secure channel** to distribute keys is **needed**.
- Notation**
 - $Y = E(K, X)$ - (encryption), K : Secret Key, X : Plaintext
 - $X = D(K, Y)$ - (decryption), K : Secret Key, Y : Ciphertext

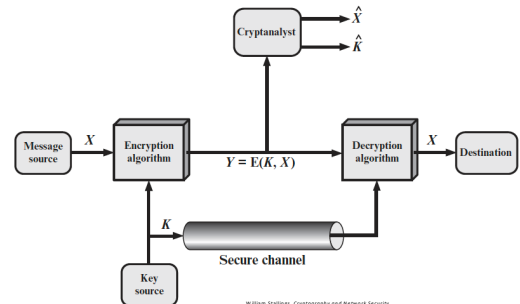
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Symmetric Cryptosystem



Sender and receiver **share the same key (secret key)**

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Attacks on Symmetric Cryptosystem

Cryptanalysis relay on

- the **nature** of the **algorithm**
- some knowledge of the general **characteristics** of the **plaintext or plaintext-ciphertext pairs**

Brute-force attack: try every possible **key** on a piece of **ciphertext** until an intelligible **translation** into plaintext is obtained.

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Attacks on Symmetric Cryptosystem

Type of Attack	Known to Cryptanalyst
Ciphertext Only	- Encryption algorithm - Ciphertext
Known Plaintext	- Encryption algorithm - Ciphertext - One or more plaintext-ciphertext pairs formed with the secret key
Chosen Plaintext	- Encryption algorithm - Ciphertext - Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	- Encryption algorithm - Ciphertext - Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	- Encryption algorithm - Ciphertext - Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key - Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

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Secure Algorithms

An algorithm is **unconditionally secure** if **ciphertext generated** by the scheme **does not contain enough information** to determine uniquely the **corresponding plaintext**, no matter how much ciphertext is available.

An **encryption** scheme is **computationally secure** if :

- The **cost of breaking** the cipher exceeds the value of the encrypted information.
- The **time required** to break the cipher exceeds the useful lifetime of the information.

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Substitution Techniques

Caesar cipher

Replace each letter of the **alphabet** with the letter standing **three places further** down the alphabet.

plain: meet me after the toga party
cipher: PHHW PH DIWHU WKH WRJD SDUWB

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Let us assign a numerical equivalent to each letter:

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

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Substitution Techniques

Caesar cipher

$$C = E(3, p) = (p + 3) \bmod 26$$

A shift may be of any amount, so that the general Caesar algorithm is

$$C = E(k, p) = (p + k) \bmod 26$$

p is plaintext letter

$$p = D(k, C) = (C - k) \bmod 26$$

Problems with Caesar cipher

1. The encryption and decryption algorithms are known
2. There are only 25 keys to try.
3. The language of the plaintext is known and easily recognizable

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Substitution Techniques

Caesar cipher

KEY	PHHW PH DIWHU WKH WRUD SDWB
1	ogvy og chvt vjg vjic rotva
2	ntu nf bqufe uif qhbb qbeur
3	meet me after the toga party
4	lode lo zeadq zga anrz ozqax
5	kccr kc ydrcp rfc rney nypw
6	jbbq jb xqgbo qeb qlde meoqv
7	laap la wqpan pda pbow jengp
8	hzso hz vaomz ocr ojkw kmot
9	gyyn gy uznyl nby niau juins
10	fxxm fx tymek max mhat itkmr
11	ewel ew xkijw lzw lqys hsjlq
12	dvdk dv rskol kpy kfar grikp
13	cusj cu qvjuh jxu jwqg fqhjo
14	btti bt puity iwt idvp epqin
15	aash as othaf hvs huoq dofhm
16	zrrg zr nagre gur gbfn enegl
17	yqfq yq mrfqd ftq faam bedfk
18	xppe xp lqpcq esp ezrl aloaj
19	wood wo kpdob dro dyqk zkbdi
20	vnnc vn jocna cqn cnpj yjaeh
21	ummb um inbes bpm bwol xizbg
22	tlla tl hamly xol avbh whyaf
23	akkr ak qlkxw xnk xung wpxce
24	rjyy rj fkyiw ymj ytif ufeyd
25	qlix ql ejxiv xli xake tewxc

If it is known that a given ciphertext is a Caesar cipher, then a brute-force cryptanalysis is easily performed: simply try all the 25 possible keys.

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Substitution Techniques

Monoalphabetic ciphers

Increase in the key space may be achieved by arbitrary substitutions

A permutation of a finite set of elements S is an ordered sequence of all the elements of S , with each element appearing exactly once.

There are $n!$ permutations of a set of n elements.

Monoalphabetic substitution cipher uses permutations of a single cipher alphabet (mapping from plain alphabet to cipher alphabet).

For example, if $S = \{a, b, c\}$, there are six permutations of S :

abc, acb, bac, bca, cab, cba

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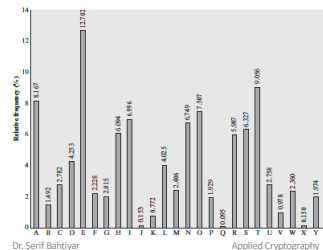
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Substitution Techniques

Monoalphabetic ciphers

Attacks on monoalphabetic ciphers depends on the nature of plaintext.

Frequency of letters is determined and compared to a standard frequency distribution of the alphabet.



Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet

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Symmetric Cryptosystem

Playfair cipher

Playfair cipher is the best-known multiple-letter encryption cipher.

Playfair cipher treats digrams in the plaintext as single units and translates these units into ciphertext digrams.

Playfair algorithm is based on the use of a 5×5 matrix of letters constructed using a keyword.

Playfair cipher is a great advance over simple monoalphabetic ciphers. For one thing, whereas there are only 26 letters, there are $26 \times 26 = 676$ digrams.

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Symmetric Cryptosystem

Playfair cipher

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

The keyword is **monarchy**. The matrix is constructed by filling in the letters of the keyword (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetic order. The letters **I** and **J** count as one letter.

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Symmetric Cryptosystem

Playfair cipher

Plaintext is encrypted **two letters** at a time, according to the following rules:

1. Repeating plaintext letters, which are in the same pair, are **separated** with a **filler letter**, such as **x**, so that **balloon** would be treated as **ba lx lo on**.
2. Two plaintext letters that fall in the **same row** of the matrix are each **replaced** by the letter to the **right**, with the **first element of the row** circularly following the last. For example, **ar** is encrypted as **RM**.
3. Two plaintext letters that fall in the **same column** are each **replaced** by the letter beneath, with the **top element of the column** circularly following the last. For example, **mu** is encrypted as **CM**.
4. Otherwise, each plaintext letter in a pair is **replaced** by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, **hs** becomes **BP** and **ea** becomes **IM** (or **JM**, as the encipherer wishes).

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Substitution Techniques

Polyalphabetic ciphers

Polyalphabetic substitution cipher use **different monoalphabetic substitutions** as one proceeds through the plaintext message.

Main features of polyalphabetic cipher:

1. A **set of related monoalphabetic** substitution rules is used.
2. A **key determines** which particular **rule** is chosen for a given transformation.

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Substitution Techniques

Polyalphabetic ciphers – Vigenere cipher

The **set** of related **monoalphabetic substitution** rules **consists of the 26 Caesar ciphers** with shifts of 0 through 25.

Each cipher is denoted by a **key letter**, which is the ciphertext letter that substitutes for the plaintext letter **a**. Thus, a **Caesar** cipher with a shift of 3 is denoted by the key value 3.

Plaintext $P = p_0, p_1, p_2, \dots, p_{n-1}$

Key $K = k_0, k_1, k_2, \dots, k_{m-1}$

Ciphertext $C = C_0, C_1, C_2, \dots, C_{n-1}$

$E(K, P) = E[(k_0, k_1, k_2, \dots, k_{m-1}), (p_0, p_1, p_2, \dots, p_{n-1})]$
 $= (p_0 + k_0) \bmod 26, (p_1 + k_1) \bmod 26, \dots, (p_{m-1} + k_{m-1}) \bmod 26,$
 $(p_m + k_0) \bmod 26, (p_{m+1} + k_1) \bmod 26, \dots, (p_{2m-1} + k_{m-1}) \bmod 26, \dots$

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Substitution Techniques

Polyalphabetic ciphers – Vigenere cipher

A general equation of the **encryption** process

$$C_i = (p_i + k_i \bmod m) \bmod 26$$

A general equation of the **decryption** process

$$p_i = (C_i - k_i \bmod m) \bmod 26$$

key: *deceptive*
 plaintext: *wearediscoveredsaveyourself*
 ciphertext: *ZTCYTWQNGRZGYWAVZHOCQYGLMGJ*

Keyword: *deceptive*

Message: *we are discovered save yourself*

Expressed numerically, we have the following result.

key	3	4	2	4	15	19	8	21	4	3	4	2	4	15
plaintext	22	4	0	17	4	3	8	18	2	14	21	4	17	4
ciphertext	25	8	2	21	19	22	16	13	6	17	25	6	21	19

key	19	8	21	4	3	4	2	4	15	19	8	21	4
plaintext	3	18	0	21	4	24	14	20	17	18	4	11	5
ciphertext	22	0	21	25	7	2	16	24	6	11	12	6	9

Multiple ciphertext letters

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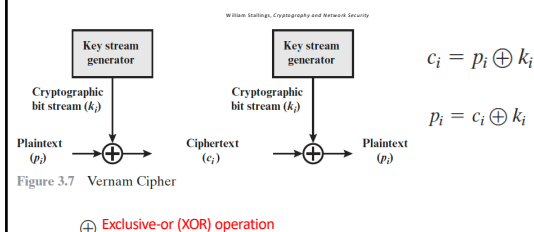
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Substitution Techniques

Polyalphabetic ciphers – Vernam cipher

The **ultimate defense** against such a **cryptanalysis** is to choose a **keyword** that is as long as the plaintext and has **no statistical relationship** to it.



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Substitution Techniques

Polyalphabetic ciphers – One time pad (OTP)

One time pad requires a **new key of the same length** as the **new message**.

The **security** of OTP is entirely due to the **randomness of the key**.

OTP is the **only cryptosystem** that exhibits what is referred to as **perfect secrecy**.

ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIURFPLUYTS

We now show two different decryptions using two different keys:

ciphertext: ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIURFPLUYTS

key: *pxlmvmsydofoyrvzwc tnlbncvqdupahfzlmnyih*

plaintext: *mr mustard with the candlestick in the hall*

ciphertext: ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIURFPLUYTS

key: *pftgpmiydgaxgoufhk1llmhsqdgogtewbqfgyovuhwt*

plaintext: *miss scarlet with the knife in the library*

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Transposition Techniques

Transposition cipher is achieved by performing **some sort of permutation** on the **plaintext** letters.

In **rail fence technique**, the **plaintext** is written down as a **sequence of diagonals** and then **read off as a sequence of rows**.

For example, to encipher the message "meet me after the toga party" with a rail fence of depth 2, we write the following:

```
m e m a t r h t g p r y
e t e f e t e o a a t
```

The encrypted message is

```
MEMATRHTGPRYETEFETEOAAT
```

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Transposition Techniques

A **more complex scheme** is to **write** the message in a **rectangle, row by row**, and **read** the message off, **column by column**, but **permute the order of the columns**. The **order of the columns** then becomes the **key** to the algorithm.

```
Key:      4 3 1 2 5 6 7
Plaintext: a t t a c k p
           o s t p o n e
           d u n t i l t
           w o a m x y z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ
```

A pure **transposition cipher** is **easily recognized** because it **has the same letter frequencies** as the original **plaintext**.

Solution: more than one stage of transposition

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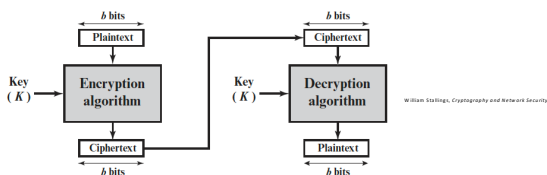
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Block ciphers

In **block cipher**, a **block of plaintext** is treated as a **whole** and it is used to produce a **ciphertext block of equal length**.



The vast **majority of network-based symmetric** cryptographic applications make use of **block ciphers**.

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Block ciphers

Fiestel cipher

A **block cipher** operates on a plaintext **block of n bits** to produce a ciphertext **block of n bits**.

A **transformation** is called **reversible**, or **nonsingular** if there are **2ⁿ** possible **different plaintext blocks**.

Reversible Mapping		Irreversible Mapping	
Plaintext	Ciphertext	Plaintext	Ciphertext
00	11	00	11
01	10	01	10
10	00	10	01
11	01	11	01

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Block ciphers

Fiestel cipher

The most general form of **block cipher** and can be used to define any **reversible mapping** between **plaintext** and **ciphertext**.

Feistel refers to this as the **ideal block cipher**, because it allows for the maximum number of possible **encryption mappings from the plaintext block**.

Small block size is a weakness in **ideal block cipher**. It allows **statistical analysis**!

Large block size is **not practical** for **implementation** and **performance**.

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Block ciphers

Fiestel cipher

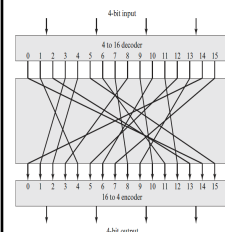


Figure 4.2 General n-bit Block Substitution (shown with n = 4)

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Table 4.1 Encryption and Decryption Tables for Substitution Cipher of Figure 4.2

Plaintext	Ciphertext	Ciphertext	Plaintext
0000	1110	0000	1110
0001	0100	0001	0011
0010	1101	0010	0100
0011	0001	0011	1000
0100	0010	0100	0001
0101	1111	0101	1000
0110	1011	0110	1010
0111	1000	0111	1111
1000	0011	1000	0111
1001	1010	1001	1101
1010	0110	1010	1001
1011	1100	1011	0110
1100	0101	1100	1011
1101	1001	1101	0010
1110	0000	1110	0000
1111	0111	1111	0101

For an **n-bit ideal block cipher**, length of the key defined is **n * 2ⁿ** bits.

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Block ciphers

Fiestel cipher

Fiestel cipher approximate the ideal block cipher by utilizing the concept of a **product cipher**, which is the execution of two or more simple ciphers in sequence.

The **product** is **cryptographically stronger** than any of the component ciphers.

This approach develops a block cipher with a **key length of k** bits and a **block length of n** bits, allowing a **total of 2^k** possible transformations, rather than the $2^n!$ transformations available with the ideal block cipher.

Fiestel uses both **substitutions** and **permutations**.

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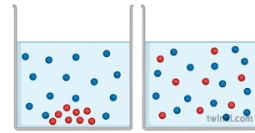
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Block ciphers

Fiestel cipher

In a **strongly ideal cipher**, all statistics of the ciphertext are **independent** of the particular **key** used.

In **diffusion**, the **statistical structure** of the **plaintext** is dissipated into **long-range** statistics of the **ciphertext**.



In a **binary block cipher**, **diffusion** is done **many permutations on data**.

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Block ciphers

Fiestel cipher

Confusion seeks to make the **relationship** between the **statistics** of the **ciphertext** and the value of the encryption **key** as complex as possible.



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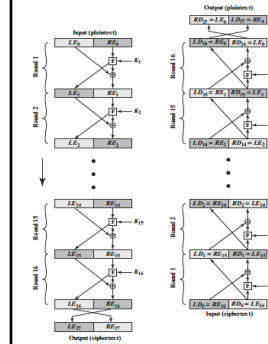
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Block ciphers

Fiestel cipher

Fiestel network depend on

- Block size
- Key size
- Number of rounds
- Subkey generation algorithm
- Round function F



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Block ciphers

Data Encryption Standard (DES)

- Most **widely used** **block cipher** in world
- Adopted in 1977 by NIST
- Encrypts 64-bit data using 56-bit key
- Has **widespread use**
- **Considerable** controversy over its **security**
- DES is basically a **product cipher**
 - several rounds of **substitutions** and **permutations**
 - actually **not** that **simple**
- Originally designed for **hardware** implementation
 - software implementations validated in 1993
 - but **software DES** is slow

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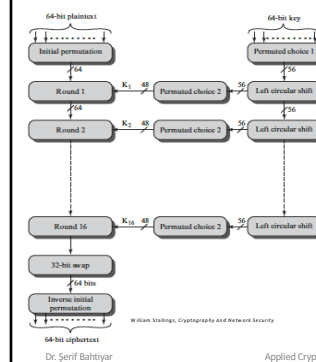
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Block ciphers

Data Encryption Standard (DES)

The **strength** of **DES**

- Key size
- Nature of the algorithm



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Avalanche Effect

Table 4.3 Avalanche Effect in DES: Change in Plaintext

Round	δ	Round	δ	
	02468aceca86420 12468aceca86420	9	c11bfce09887f6c6c 99f911532eed7d94	32
1	3cf03cf0bfed2845 3cf03cf0bfed2845	10	887bf6e4d0f07f8f 2eed7d400f023094	31
2	bad2284559be723 bad3284539be7a3	11	6007f8ebf596506 d02304a455da9c4	37
3	999b723020be39e 39a79a3131cbb83	18	5298506a738538a 45da9ca47f6ef3cf3	3
4	0bae3b9ed421549 71cbb3cbccaca55e	13	73853b8bf6af2c4e 7f6a3c23bclaa29d	4
5	424156491bb3a41 ccaca55ed1ce3653	14	c6a26e4eb4b0b75 40d90107d409	33
6	18bf3af109f16623 d16c3653cf402c68	15	56b0b07575ebfd8f 1e07d0a30e0e6dc	30
7	9616fe2367117f2 cf402c682020cfbc	16	78e58dfc2589490 1ce26da655e5f59	32
8	67117cf21b2cf019 2b26cfce99f91153	33	05d0e23b89acc3a da07d97f683f2a	32

A desirable property of any encryption algorithm is that a small change in either the plaintext or the key should produce a significant change in the ciphertext.

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Block ciphers

Design Principles

- Number of rounds
 - The **greater the number of rounds**, the more **difficult** it is to perform **cryptanalysis**, even for a relatively weak.
- Design of function **F**
 - The **heart** of a Feistel block cipher is the function **F**, which provides the element of confusion in a Feistel cipher.
 - **F** be non-linear
- Key schedule algorithm

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Random Numbers

Random numbers play an important role in the use of encryption for various network security applications.

Random numbers in cryptography are used for

- **Nonces** in authentication protocols to prevent replay attacks
- **Session** keys, symmetric keys
- **Public key** generation
- **Keystream** for stream ciphers
- **Key distribution** scenarios, such as Kerberos (prevents replay attacks)

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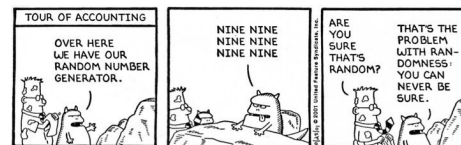
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Random Numbers

- **Characteristics** of random numbers
 - **Statistical** randomness criteria
 - **Uniform distribution** of zeros and ones
 - **Independence** of the bits in the sequence
 - **Unpredictability** of future values from previous values



- True random numbers provide these but very hard to obtain and use in practice

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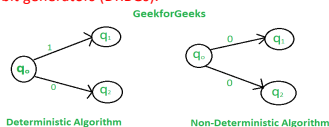
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Random Numbers

Two different strategies for generating random numbers

Deterministic algorithms

- Known as **pseudorandom number generators (PRNGs)** or **deterministic random bit generators (DRBGs)**.



Non-deterministic algorithms

- use some **physical source** that produces some **sort of random output**.
- **true random number generators (TRNGs)** or **non-deterministic random bit generators (NRBGs)**

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Pseudorandom Numbers

- Pseudorandom numbers are generated by deterministic algorithms and are not statically random.
- May pass many tests of randomness.
- Under most circumstances, pseudorandom numbers perform as well as true random numbers.

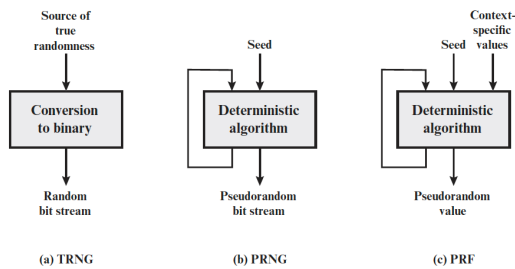
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Pseudorandom Numbers



TRNG = true random number generator
 PRNG = pseudorandom number generator
 PRF = pseudorandom function

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Pseudorandom Numbers

- True random number generator (TRNG) takes as input a source that is effectively random; the source is often referred to as an entropy source.
- PRNG takes as input a fixed value, called the seed, and produces a sequence of output bits using a deterministic algorithm. An algorithm that is used to produce an open-ended sequence of bits.
- Pseudorandom function (PRF) is used to produce a pseudorandom string of bits of some fixed length.

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Pseudorandom Numbers

PRNG Requirements

- The basic requirement: an adversary who does not know the seed is unable to determine the pseudorandom string.
- Randomness
 - Uniformity
 - Scalability
 - Consistency
- Unpredictability
 - Forward unpredictability
 - Backward unpredictability
- Seed Requirements

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Pseudorandom Numbers

PRNG Design Algorithms

- Purpose-built algorithms
- Algorithms based on existing cryptographic algorithms
- Symmetric block ciphers
- Asymmetric ciphers
- Hash functions and message authentication codes



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Pseudorandom Numbers

A set of PRNGs: Linear Congruential Generators

The algorithm is parameterized with four numbers, as follows:

m	the modulus	$m > 0$
a	the multiplier	$0 < a < m$
c	the increment	$0 \leq c < m$
X_0	the starting value, or seed	$0 \leq X_0 < m$

The sequence of random numbers $\{X_n\}$ is obtained via the following iterative equation:

$$X_{n+1} = (aX_n + c) \bmod m$$

If m , a , c , and X_0 are integers, then this technique will produce a sequence of integers with each integer in the range $0 \leq X_n < m$.

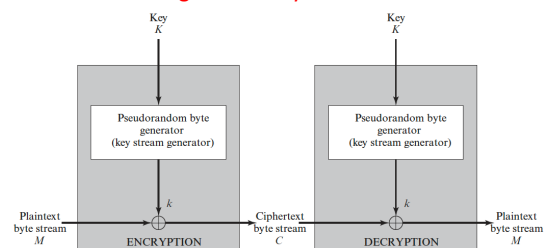
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Stream Ciphers

Stream cipher encrypts plaintext one byte at a time, one bit at a time or on units larger than a byte at a time.



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Stream Ciphers

Key is input to a pseudorandom bit generator that produces a stream of 8-bit numbers that are apparently random.

Output of the generator, called a keystream, is combined one byte at a time with the plaintext stream using the bitwise exclusive-OR (XOR) operation.

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Stream Ciphers

Ex:

If the next byte generated by the generator is 01101100 and the next plaintext byte is 11001100.

Encryption

```
11001100 plaintext
⊕ 01101100 key stream
10100000 ciphertext
```

Decryption

```
10100000 ciphertext
⊕ 01101100 key stream
11001100 plaintext
```

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Stream Ciphers

Design principles of stream ciphers

1. The encryption sequence should have a large period.
2. The keystream should approximate the properties of a true random number stream as close as possible.
3. To guard against brute-force attacks, the key needs to be sufficiently long.

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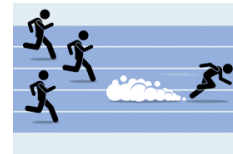
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Stream Ciphers

- With a properly designed pseudorandom number generator, a stream cipher can be as secure as a block cipher of comparable key length.



- Stream ciphers that do not use block ciphers as a building block are typically faster and use far less code than do block ciphers.

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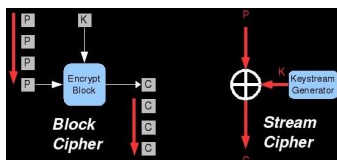
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Stream Ciphers

One advantage of a block cipher is that you can reuse keys.



For applications that require encryption/decryption of a stream of data, such as over a data communications channel or a browser/Web link, a stream cipher might be the better alternative.

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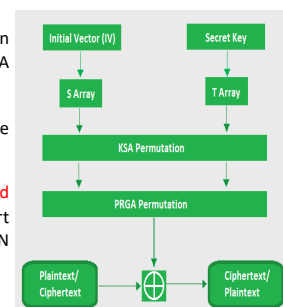
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RC4

- RC4 is a stream cipher designed in 1987 by Ron Rivest for RSA Security.

- The algorithm is based on the use of a random permutation.

- RC4 is used in the WiFi Protected Access (WPA) protocol that are part of the IEEE 802.11 wireless LAN standard.



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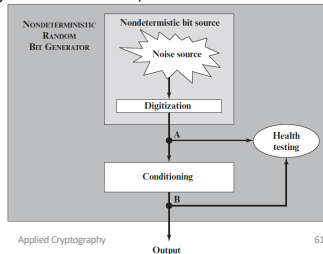
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True Random Generators

TRNG uses a **nondeterministic source** to produce **randomness**.

Operates by **measuring unpredictable natural processes**,

- pulse detectors of ionizing radiation events,
- gas discharge tubes,
- leaky capacitors.



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Introduction to Public-key Crypto

- **Public-key cryptography** provides a **radical** departure from all cryptosystems has gone before.

$$f(x)$$

- **Public-key algorithms** are based on **mathematical functions** rather than on **simple operations** on bit patterns.

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History of Public-key Cryptography

- **Public-key cryptography** is **invented** by Whitfield Diffie and Martin Hellman in 1976
- **NSA** says that they knew public-key cryptography back in **60's**
- **First documented** introduction of public-key cryptography is by **James Ellis** of UK's Communications-Electronics Security Group in 1970
- **RSA**: Block cipher in which the plaintext are integers between 0 and $n-1$ for some n .

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Some Misconceptions

- **Public-key cryptography** **replaces symmetric** cryptography
- **Public-key cryptography** is **more secure** (no evidence for that, security mostly depends on the key size in both schemes)



- **Key distribution** is trivial in **public-key cryptography** since public keys are public (key distribution is **easier**, but **not trivial**)

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Symmetric Cryptography Problems

Public-key cryptography (PKC) evolved from **two of the most difficult problems** of **symmetric encryption**:

Key distribution

- Symmetric crypto **requires a trusted Key Distribution Center (KDC)**
- In symmetric crypto two **communicants share a key**, which somehow has been distributed to them



- In PKC you do **not need a KDC** to distribute secret keys, but you still need trusted third parties

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Symmetric Cryptography Problems

Digital signatures (non-repudiation)

- **Not possible** with symmetric crypto
- **Broader requirement** than authentication



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Application Categories

- **Encryption/decryption** : to provide **secrecy**
- **Digital signatures** : to provide **authentication** and
- **Key exchange**: to **agree** on a session key



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Some Terminology

Asymmetric Keys

Two related keys, a public key and a private key, that are used to perform complementary operations, such as encryption and decryption or signature generation and signature verification.

Public Key Certificate

A digital document issued and digitally signed by the private key of a Certification Authority that binds the name of a subscriber to a public key. The certificate indicates that the subscriber identified in the certificate has sole control and access to the corresponding private key.

Public Key (Asymmetric) Cryptographic Algorithm

A cryptographic algorithm that uses two related keys, a public key and a private key. The two keys have the property that deriving the private key from the public key is computationally infeasible.

Public Key Infrastructure (PKI)

A set of policies, processes, server platforms, software and workstations used for the purpose of administering certificates and public-private key pairs, including the ability to issue, maintain, and revoke public key certificates.

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Conventional vs. Public-key Encryption

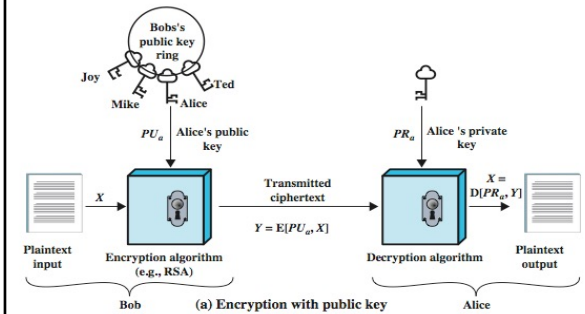
Conventional Encryption	Public-Key Encryption
Needed to Work: 1. The same algorithm with the same key is used for encryption and decryption. 2. The sender and receiver must share the algorithm and the key. Needed for Security: 1. The key must be kept secret. 2. It must be impossible or at least impractical to decipher a message if the key is kept secret. 3. Knowledge of the algorithm plus samples of ciphertext must be insufficient to determine the key.	Needed to Work: 1. One algorithm is used for encryption and a related algorithm for decryption with a pair of keys, one for encryption and one for decryption. 2. The sender and receiver must each have one of the matched pair of keys (not the same one). Needed for Security: 1. One of the two keys must be kept secret. 2. It must be impossible or at least impractical to decipher a message if one of the keys is kept secret. 3. Knowledge of the algorithm plus one of the keys plus samples of ciphertext must be insufficient to determine the other key.

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Principles



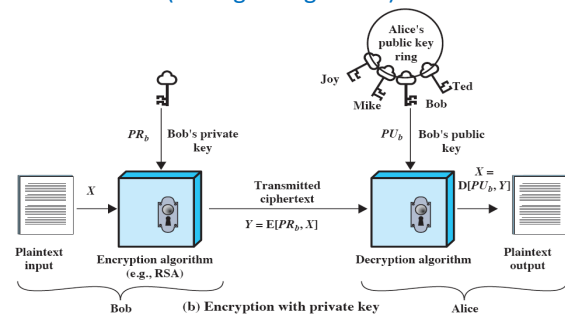
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Principles

Authentication (for Digital Signature)



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Principles

With the message X and the encryption key PU_b as input, A forms the ciphertext $Y = [Y_1, Y_2, \dots, Y_N]$:

$$Y = E(PU_b, X)$$

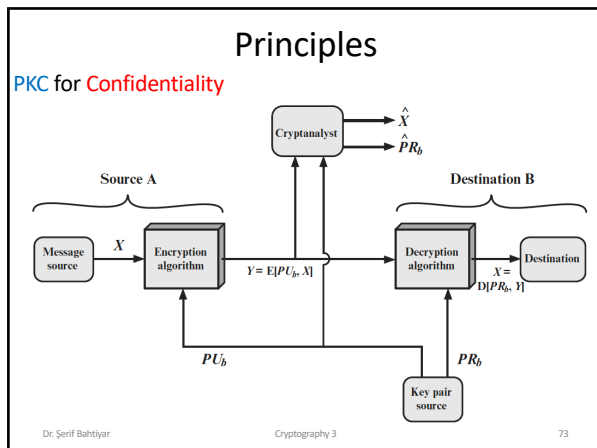
The intended receiver, in possession of the matching private key, is able to invert the transformation:

$$X = D(PR_b, Y)$$

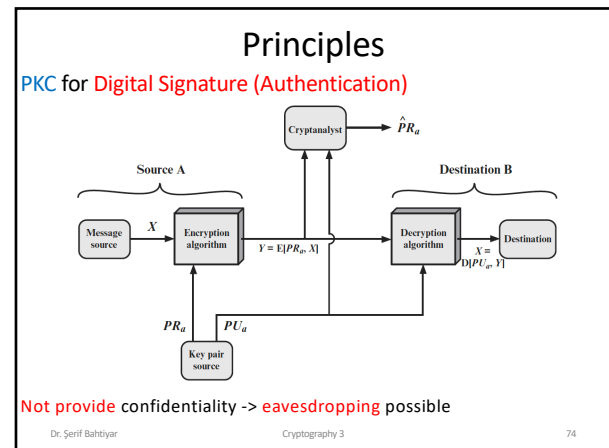
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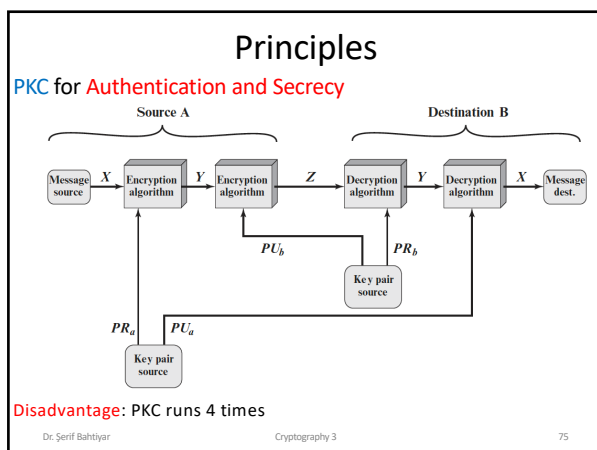
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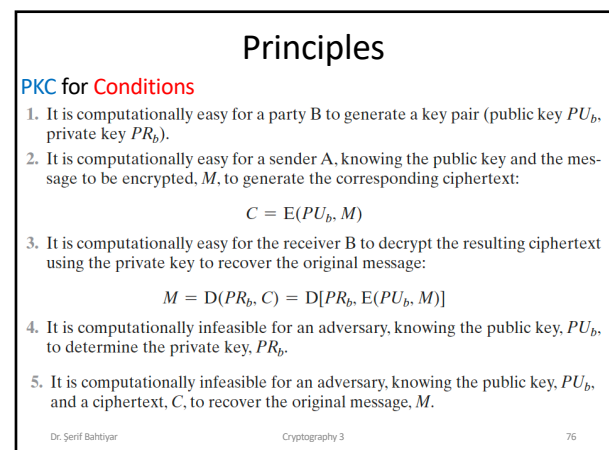
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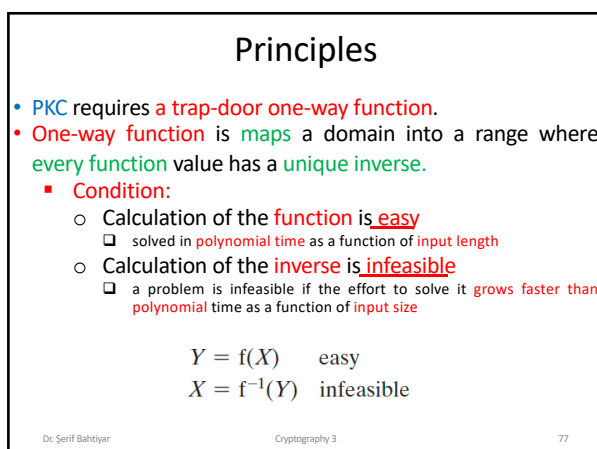
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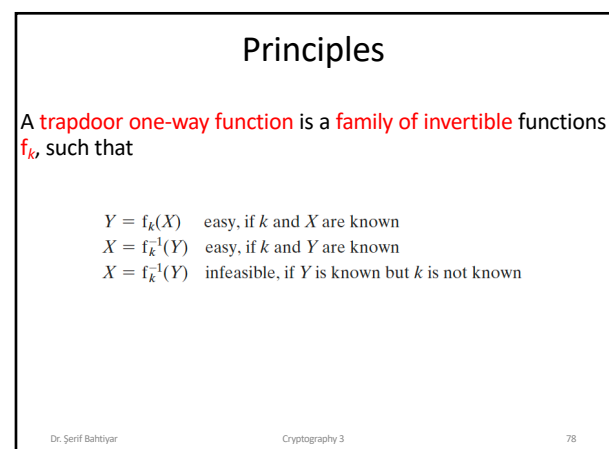
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RSA Algorithm

Rivest-Shamir-Adleman (RSA) scheme has been the **most widely accepted and implemented** general-purpose approach to public-key encryption.

RSA

RSA scheme is a cipher in which the **plaintext** and **ciphertext** are **integers** between **0** and **$n - 1$** for some **n** .

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RSA Algorithm

- RSA makes use of an expression with **exponentials**. **Plaintext** is encrypted in **blocks**, with **each block** having a **binary value less** than some number **n** .

- For **plaintext** block **M** and **ciphertext** block **C** :

$$C = M^e \bmod n$$

$$M = C^d \bmod n = (M^e)^d \bmod n = M^{ed} \bmod n$$

- Both **sender** and **receiver** know value of **n** . **Sender** knows value of **e** , and only **receiver** knows value of **d** .

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RSA Algorithm

Requirements of RSA

- It is possible to find values of e, d , and n such that $M^{ed} \bmod n = M$ for all $M < n$.
- It is relatively easy to calculate $M^e \bmod n$ and $C^d \bmod n$ for all values of $M < n$.
- It is infeasible to determine d given e and n .

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RSA Algorithm

Key Generation

Select p, q	p and q both prime, $p \neq q$
Calculate $n = p \times q$	
Calculate $\phi(n) = (p - 1)(q - 1)$	
Select integer e	$\gcd(\phi(n), e) = 1; 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \pmod{\phi(n)}$
Public key	$PU = \{e, n\}$
Private key	$PR = \{d, n\}$

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RSA Algorithm

Encryption

Plaintext:	$M < n$
Ciphertext:	$C = M^e \bmod n$

Decryption

Ciphertext:	C
Plaintext:	$M = C^d \bmod n$

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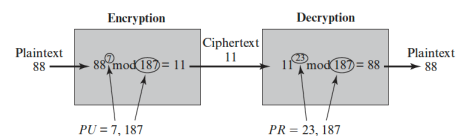
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RSA Algorithm

Example

- Select two prime numbers, $p = 17$ and $q = 11$.
- Calculate $n = pq = 17 \times 11 = 187$.
- Calculate $\phi(n) = (p - 1)(q - 1) = 16 \times 10 = 160$.
- Select e such that e is relatively prime to $\phi(n) = 160$ and less than $\phi(n)$; we choose $e = 7$.
- Determine d such that $de \equiv 1 \pmod{160}$ and $d < 160$. The correct value is $d = 23$, because $23 \times 7 = 161 = (1 \times 160) + 1$; d can be calculated using



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RSA Algorithm

Example Cont.

The resulting keys are public key $PU = \{7, 187\}$ and private key $PR = \{23, 187\}$. The example shows the use of these keys for a plaintext input of $M = 88$. For encryption, we need to calculate $C = 88^7 \bmod 187$. Exploiting the properties of modular arithmetic, we can do this as follows.

$$88^7 \bmod 187 = [(88^4 \bmod 187) \times (88^2 \bmod 187) \times (88^1 \bmod 187)] \bmod 187$$

$$88^1 \bmod 187 = 88$$

$$88^2 \bmod 187 = 7744 \bmod 187 = 77$$

$$88^4 \bmod 187 = 59,969,536 \bmod 187 = 132$$

$$88^7 \bmod 187 = (88 \times 77 \times 132) \bmod 187 = 894,432 \bmod 187 = 11$$

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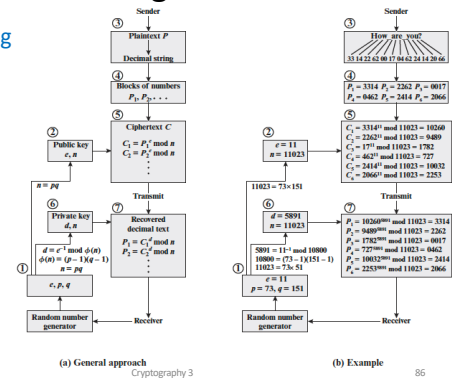
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RSA Algorithm

RSA Processing



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Summary

- Basic concepts of cryptography
- Classification of cryptographic systems
- Symmetric Cryptosystem
- Block cipher
- Fiestel cipher
- Design Principles
- Random numbers for cryptography
- Stream ciphers
- Conventional vs. Public-key Encryption
- Public-key cryptography details

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Questions?

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